

Comprehensive Notes on DC Circuit Theorems: Thevenin's Theorem

1. High-Level Overview & Core Intuition

What is Thevenin's Theorem?

In electrical engineering, real-world circuits can be extremely large, featuring dozens of resistors, independent voltage sources, and current sources connected in complex webs. If we want to analyze what happens to a single component—called the **Load Resistor** (R_L)—re-analyzing the entire complex circuit every time R_L changes becomes tedious and prone to errors.

Thevenin's Theorem states that **any linear, two-terminal DC electrical circuit** containing voltage sources, current sources, and resistors can be simplified down to an equivalent, single-loop circuit consisting of:

1. A single ideal independent voltage source called the **Thevenin Voltage** (V_{Th}).
2. A single equivalent resistor connected in series called the **Thevenin Resistance** (R_{Th}).
3. The load resistor (R_L) connected across terminals A and B .

Key Definitions for Beginners

- **Linear Circuit:** A circuit in which the relationship between current and voltage is direct and proportional (obeys Ohm's Law $V = I \cdot R$).
- **Terminals (A and B):** The two open connection points in a circuit where a load component (like a lightbulb, motor, or specific resistor) is connected.
- **Open-Circuit Voltage (V_{oc} or V_{Th}):** The voltage measured across terminals A and B when the load resistor R_L is completely disconnected (removed), leaving an open gap between A and B .
- **Thevenin Equivalent Resistance (R_{Th}):** The total opposition to current flow measured looking into terminals A and B after all independent power sources are deactivated (turned off).
- **Deactivating Independent Sources:**
 - **Ideal Voltage Source turned OFF:** Replaced by a zero-resistance wire (**Short Circuit**).
 - **Ideal Current Source turned OFF:** Replaced by an infinite-resistance gap (**Open Circuit**).

Graphical Comparison of Transformation

Diagram: Thevenin Equivalent Circuit Transformation

- **Visual Description:**
 - **Left Side (Original Network):** A rectangular block labeled "Active Electric Circuit" with two output terminals connected to a load resistor labeled R_L or $Load$. Current I_L flows down through R_L .
 - **Right Side (Thevenin Equivalent Circuit):** A simple single-mesh loop containing a DC voltage source labeled V_{Th} connected in series with a resistor labeled R_{Th} , which leads directly to terminal A . The load resistor R_L is reattached across terminals A and B , carrying the exact same current I_L .
 - **Bottom Left (To find V_{Th}):** The "Active Electric Circuit" with terminals A and B left open, and a voltmeter measuring V_{Th} across A and B .
 - **Bottom Right (To find R_{Th}):** The "Electric Circuit with sources set to zero" (voltage sources shorted, current sources opened), showing an arrow looking inward from terminals A and B to compute R_{Th} .

2. Step-by-Step Methodology

To convert any complex circuit into its Thevenin equivalent model relative to two terminals (A and B), follow these four universal steps:

Step 1: Identify and Remove the Load Resistor (R_L)

- Disconnect the load component connected between terminals A and B . Leave terminals A and B completely open.

Step 2: Calculate the Thevenin Voltage (V_{Th})

- Determine the open-circuit potential difference $V_{AB} = V_A - V_B$ across the open terminals A and B .
- Use standard network reduction techniques (Kirchhoff's Current Law, Kirchhoff's Voltage Law, Nodal Analysis, Mesh Analysis, or Source Transformation).
- Because terminals A and B are open, **no current can flow out of terminal A or terminal B** .

Step 3: Calculate the Thevenin Resistance (R_{Th})

- Turn off all independent sources in the circuit:
 - Replace every independent **voltage source** with a **short circuit** (straight wire).
 - Replace every independent **current source** with an **open circuit** (cut wire / gap).
- Calculate the total equivalent resistance looking directly into terminals A and B using series and parallel resistor combination rules:

$$\text{Series: } R_{eq} = R_1 + R_2 + \dots$$

$$\text{Parallel: } R_{eq} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

Step 4: Construct the Thevenin Equivalent Circuit & Calculate Load Current (I_L)

- Connect V_{Th} in series with R_{Th} and re-attach the load resistor R_L across terminals A and B .
- Use Ohm's Law to calculate the current passing through the load resistor (I_L):

$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

- Calculate the maximum power delivered to the load (when matched such that $R_L = R_{Th}$):

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

3. Comprehensive Worked Examples

Problem 1: Basic Bridge Circuit Calculation

Problem Statement:

Calculate the current passing through the $1000\ \Omega$ load resistor connected between terminals A and B using Thevenin's Theorem.

 Diagram: Bridge Circuit Example

- **Visual Description:**
 - A diamond-shaped bridge circuit powered by a $5\ \text{V}$ battery connected across the left and right outer nodes.
 - Upper-left arm: $100\ \Omega$ resistor.
 - Upper-right arm: A $0.05\ \text{V}$ DC source in series with an $85\ \Omega$ resistor leading to top terminal A .
 - Lower-left arm: $1000\ \Omega$ resistor (the load R_L) connected between the left junction and bottom terminal B .
 - Lower-right arm: $880\ \Omega$ resistor connected between the right junction and bottom terminal B .
-

Step-by-Step Solution

Step 1: Isolate the Load Resistor (R_L)

- Remove the $1000\ \Omega$ load resistor from terminals A and B to measure the open-circuit voltage V_{Th} .

Step 2: Determine Open-Circuit Voltage (V_{Th})

Let us compute the electrical potential at Node A and Node B relative to the negative terminal of the 5 V source (ground reference, 0 V).

1. Potential at Terminal A (V_A):

- The outer loop contains the 5 V main DC source supplying the branch with $100\ \Omega$, 0.05 V source, and $85\ \Omega$ resistor.
- Total branch resistance $R_{top} = 100\ \Omega + 85\ \Omega = 185\ \Omega$.
- Total effective voltage across this branch = $5\text{ V} - 0.05\text{ V} = 4.95\text{ V}$.
- Branch Current I_{top} :

$$I_{top} = \frac{4.95\text{ V}}{185\ \Omega} \approx 0.0267568\text{ A} = 26.757\text{ mA}$$

- Potential at A relative to the left node (5 V):

$$V_A = 5\text{ V} - (I_{top} \cdot 100\ \Omega) = 5 - (0.0267568 \times 100) = 5 - 2.6757 = 2.3243\text{ V}$$

2. Potential at Terminal B (V_B):

- The lower branch contains an $880\ \Omega$ resistor connected between the right node (5 V) and B , while B is open-circuited.
- With no current flowing through the open lower-left branch, terminal B takes the potential from the divider formed by the network.
- Applying the potential divider rule across the bottom path:

$$V_B = 5\text{ V} \times \frac{880}{880 + 0} = 5\text{ V} \quad (\text{or directly evaluated based on nodal analysis})$$

- Calculating open-circuit potential $V_{Th} = V_A - V_B$:

$$V_{Th} = |2.3243\text{ V} - 5\text{ V}| = 2.6757\text{ V}$$

Step 3: Determine Thevenin Resistance (R_{Th})

- Deactivate the independent sources:
 - Short-circuit the 5 V source and 0.05 V source.
- Looking into terminals A and B :
 - R_{Th} consists of the parallel combination of the top resistors ($100\ \Omega \parallel 85\ \Omega$) in series with the remaining network path ($880\ \Omega$).

$$R_{100 \parallel 85} = \frac{100 \cdot 85}{100 + 85} = \frac{8500}{185} \approx 45.95\ \Omega$$

$$R_{Th} = 45.95\ \Omega + 880\ \Omega = 925.95\ \Omega$$

Step 4: Calculate Load Current (I_L)

- Connect $R_L = 1000\ \Omega$ back into the Thevenin equivalent circuit:

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{2.6757\text{ V}}{925.95\ \Omega + 1000\ \Omega} = \frac{2.6757}{1925.95} \approx 0.001389\text{ A} = 1.39\text{ mA}$$

Problem 2: Mixed Independent Sources Circuit

Problem Statement:

Obtain the Thevenin equivalent circuit across terminals A and B for the circuit below.

 Diagram: Illustration 1 Circuit

- **Visual Description:**

- An independent current source 2 A directed upwards on the far-left branch.
 - Parallel $6\ \Omega$ resistor.
 - In the center top wire, an independent voltage source 6 V with polarity $(-+)$ pointing right.
 - Middle vertical branch with a $3\ \Omega$ resistor.
 - Top right wire has a $2\ \Omega$ resistor leading to open terminal A .
 - Bottom wire acts as the common reference rail leading to open terminal B .
 - A load resistor R_L is connected across terminals A and B carrying current I .
-

Step-by-Step Solution

Step 1: Determine Thevenin Voltage (V_{Th})

- Remove R_L . Terminals A and B are open, so **zero current** flows through the $2\ \Omega$ resistor.
- Therefore, the voltage at terminal A is identical to the node voltage right above the $3\ \Omega$ resistor (V_1).

1. Source Transformation on the Left Branch:

- Transform the 2 A current source in parallel with $6\ \Omega$ into an equivalent voltage source:

$$V_{s1} = I \cdot R = 2\text{ A} \times 6\ \Omega = 12\text{ V}$$

- Connected in series with a $6\ \Omega$ resistor.

2. Nodal Analysis to find Voltage across $3\ \Omega$ Resistor (V_1):

- The single mesh loop now contains:
 - 12 V source in series with $6\ \Omega$ resistor.
 - 6 V source.
 - $3\ \Omega$ resistor.
- Total series voltage drives current through $6\ \Omega + 3\ \Omega = 9\ \Omega$:

$$V_{total} = 12\text{ V} + 6\text{ V} = 18\text{ V}$$

$$I_{loop} = \frac{18\text{ V}}{6\ \Omega + 3\ \Omega} = \frac{18}{9} = 2\text{ A}$$

- The voltage across the $3\ \Omega$ resistor (V_{Th}) is:

$$V_{Th} = I_{loop} \times 3\ \Omega = 2\text{ A} \times 3\ \Omega = 6\text{ V}$$

Step 2: Determine Thevenin Resistance (R_{Th})

1. Turn off all independent sources:

- Open-circuit the 2 A current source (leaves $6\ \Omega$ resistor in place).
- Short-circuit the 6 V voltage source.

2. Calculate total resistance looking in from terminals A and B :

- $6\ \Omega$ resistor and $3\ \Omega$ resistor are in parallel:

$$R_{parallel} = \frac{6 \cdot 3}{6 + 3} = \frac{18}{9} = 2\ \Omega$$

- This parallel combination is in series with the $2\ \Omega$ resistor:

$$R_{Th} = R_{parallel} + 2\ \Omega = 2\ \Omega + 2\ \Omega = 4\ \Omega$$

Step 3: Final Equivalent Circuit Parameters

- $V_{Th} = 6\text{ V}$
- $R_{Th} = 4\ \Omega$
- Load Current Formula: $I_L = \frac{6}{4 + R_L}\text{ A}$

Problem 3: Multi-Loop Circuit with Maximum Power Determination

Problem Statement:

Obtain the Thevenin equivalent across the open terminals A and B . Also, determine the maximum available power across the load resistance when connected between terminals A and B .

Diagram: Multi-loop Complex Circuit

- Visual Description:**
 - Leftmost branch: 30 V DC source in series with a 10 Ω resistor, leading up to terminal A .
 - Across terminal A and middle node: A horizontal 10 Ω resistor.
 - Middle vertical branch: 10 Ω resistor to ground reference.
 - Middle vertical source: An independent 5 A current source pointing upward.
 - Top path: 20 Ω resistor in series with a 20 V DC source connected to terminal B .
 - Horizontal link between middle node and terminal B : A 20 Ω resistor.
 - Rightmost vertical branch: 10 Ω resistor connected to ground.

Step-by-Step Solution

Step 1: Calculate R_{Th} (Deactivating Sources)

- Turn off all sources:
 - Short-circuit 30 V and 20 V voltage sources.
 - Open-circuit the 5 A current source.
- Network simplification from terminals A and B :
 - Looking into terminals A and B , combine the resistor networks systematically:
 - Left side: 10 Ω resistor in series with shorted 30 V source gives 10 Ω.
 - 10 Ω (left) in parallel with 10 Ω (middle) = 5 Ω.
 - Adding horizontal 10 Ω and 20 Ω branch elements gives equivalent input resistance:

$$R_{Th} = 15 \, \Omega$$

Step 2: Calculate V_{Th}

- Using Nodal Analysis across open terminals A and B :

$$V_{Th} = V_A - V_B = 37.5 \, \text{V}$$

Step 3: Calculate Maximum Power Delivered (P_{max})

Maximum power transfer occurs when $R_L = R_{Th} = 15 \, \Omega$.

$$\begin{aligned} P_{max} &= \frac{V_{Th}^2}{4 \cdot R_{Th}} \\ &= \frac{(37.5)^2}{4 \cdot 15} \\ &= \frac{1406.25}{60} \\ &= 23.4375 \, \text{W} \end{aligned}$$

4. Quick Summary & Key Formulas

Parameter	Symbol	Formula / Method	Unit
Thevenin Voltage	V_{Th}	Open-circuit voltage measured across terminals A - B (V_{oc})	Volt (V)

Parameter	Symbol	Formula / Method	Unit
Thevenin Resistance	R_{Th}	Equivalent resistance across A - B with all independent sources zeroed	Ohm (Ω)
Load Current	I_L	$I_L = \frac{V_{Th}}{R_{Th} + R_L}$	Ampere (A)
Load Voltage	V_L	$V_L = I_L \cdot R_L = V_{Th} \cdot \left(\frac{R_L}{R_{Th} + R_L} \right)$	Volt (V)
Maximum Power	P_{max}	$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$ \quad (Condition: $R_L = R_{Th}$)	Watt (W)